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An approach based on data mining and genetic algorithm to optimizing time series clustering for efficient segmentation of customer behavior

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ABSTRACT

In today's highly competitive market, organizations face significant challenges in accurately understanding and segmenting customer behavior due to the inherently dynamic and evolving nature of customer interactions over time. Traditional customer segmentation methods often neglect these temporal variations, leading to ineffective business strategies and missed opportunities. This research addresses this critical gap by introducing an innovative time series-based approach for customer behavior segmentation. By modeling each customer's behavior as a time series capturing key metrics such as purchase frequency, transaction amounts, and customer lifecycle costs the proposed method dynamically adapts to behavioral changes over time. To enhance segmentation precision, a genetic algorithm is employed to optimize feature weights, ensuring that the most relevant factors are emphasized. These optimized features are then clustered using spectral clustering to identify distinct and meaningful customer segments. The effectiveness of the proposed method is validated using 30 months of transactional data from a payment services company. The results demonstrate that the proposed approach, particularly when combined with spectral clustering and optimally weighted features, significantly surpassing the performance of traditional static segmentation techniques. This research not only provides a more accurate framework for uncovering hidden patterns in customer behavior but also delivers actionable insights for targeted marketing and personalized customer strategies.

1. Introduction

Nowadays, customer relationship management has become extremely important due to intense competition among companies in various industries (Kumar & Reinartz, 2018). With advancements in information and communication technologies, a large volume of data about customers is available to organizations. To utilize this data for strategic decision-making, data mining techniques have emerged as powerful tools for data analysis and knowledge creation (Parvaneh et al., 2014). This segmentation helps organizations interact more efficiently with customers by leveraging data analysis methods.

In this paper, we present a new model for customer segmentation using data mining techniques. This model, using appropriate data mining algorithms and methods tailored to the organization's dataset, can segment customers into different groups based on common features. In the proposed model, the features of purchase novelty, number of purchases, purchase amount, and customer cost are extracted from customer transaction data. The proposed model uses powerful clustering algorithms, including hierarchical clustering, spectral clustering, fuzzy C-means clustering, and K-means clustering, to achieve the best clustering results (Akhondzadeh-Noughabi & Albadvi, 2015; Seret et al., 2014; Yanovitzky & VanLear, 2008).

The biggest limitation of static segmentation methods is that these methods are not able to model the dynamic behavior of customers and discover meaningful patterns and trends (Khajvand & Tarokh, 2011; SARI et al., 2016). These methods are more descriptive and cannot predict the future behavior of customers. In this model, time series are used to record customer behavior and maintain the chronological order of observations. First, the features of purchase recency, purchase number, purchase amount, customer cost was extracted from customer transaction data, then customers were segmented using time series clustering. In this research, the cost has been investigated as one of the important features, which has not been sufficiently considered in previous researches. Considering cost as one of the important features in customer analysis and business management is very important. In various business fields, including banking, retail, services, etc., costs can be one of the determining factors in decisions. In the banking industry, an accurate understanding of the cost-benefit ratio for each customer

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can help the bank to provide appropriate strategies to attract, retain and upgrade customers. By knowing this ratio for each customer, the bank can evaluate the improvement of its financial performance, establish purposefulness in the use of financial resources, and realize the improvement of relations with customers who have a better cost-benefit ratio. In addition, in the existing research, multivariate time series have been used instead of univariate time series, and instead of examining the effect of individual characteristics, they have also examined their simultaneous effect together. By considering multivariate time series, it is possible to examine the simultaneous effect of different characteristics on customer behavior. This makes it possible to understand the connections and interactions between attributes and to analyze their simultaneous influence on customer behavior. This more comprehensive analysis makes it possible to discover hidden patterns and more complex relationships that are not visible in the analysis of individual features. As a result, more accurate and reliable predictions about the needs and desires of customers can be reached.

In this paper, a model is presented that displays the behavior of each customer as a time sequence of the variables of purchase novelty, number of purchases, purchase amount, and customer cost, considering the time dimension of customer behavior. Then, using the genetic algorithm, optimal weights are found for each feature, and customers are segmented with clustering algorithms. To demonstrate the utility of this model, a case study on the customers of a banking payment service company is conducted.

The structure of the paper is as follows: Section 2 reviews related works. The various techniques used in the paper and the proposed method for customer behavior analysis are presented in Section 3. The results of the proposed framework are reported in Section 4. Section 5 is dedicated to evaluating the performance of the proposed algorithm. Finally, Section 6 concludes the paper and suggests some directions for future research.

2. Literature review

In (SARI et al., 2016), the authors reviewed customer segmentation methods and highlighted that demographic segmentation (age, gender, education, occupation, income) helps in understanding customer behavior and optimizing marketing costs. In (Khajvand & Tarokh, 2011), a banking study segmented customers based on new purchases, purchase frequency, and amount, predicting each segment using time series analysis. In (Khajvand et al., 2011), two methods for customer segmentation and lifetime value calculation were presented, showing that adding item count as a new parameter had no significant effect on clustering.

In (Heldt et al.), a model for purchase frequency and amount per product showed that product data provides useful insights for marketing asset management and reduces customer value prediction errors. In (Anitha & Patil, 2019), a study using new purchase frequency and amount in retail employed K-means clustering, evaluated with the silhouette index. In (Daneshvar et al., 2020), a new multi-criteria clustering method with bi-phase optimization was introduced, enhancing the genetic algorithm with heuristic mutation for effective yet time-consuming clustering.

In (Tavakoli et al., 2018), a hybrid model combining new purchase frequency, amount, and time series was proposed, showing improved customer analysis and strategic decision-making through Short Message Service (SMS) campaigns. In (ABBASIMEHR & BAHRINI, 2022), advanced clustering and time series clustering considered the temporal dimension of customer behavior, using transaction data for new purchase frequency and amount features. In (Christy et al., 2018), new purchase frequency and amount analysis on transaction data, and clustering with K-means and fuzzy C-means, introduced a new idea for initial cluster center selection.

In (Çavdar & Ferhatosmanoğlu, 2018), an airline industry model estimated customer lifetime value using flight data and social network data, showing improved accuracy with social factors. In (Emami & Derakhshan, 2015), a company's customers were analyzed using new purchase frequency and amount, employing fuzzy clustering and customer portfolio analysis. The profitability index validated the results, categorizing customers into three clusters: superstar, regular, and dormant based on lifetime value.

In (Sivaguru, 2023), a dynamic customer segmentation (DCS) framework is introduced, comprising three phases: the first phase involves using the modified fuzzy c-means (MdFCM) algorithm to cluster new data and identify changes in cluster structures; the second phase classifies clusters based on the RFM (Recency, Frequency, Monetary) pattern; and the third phase formulates marketing strategies based on identified changes in the clusters. The MdFCM algorithm calculates distances between cluster centers, selects the minimum distance, and compares new data distances with this minimum. If the new data distance exceeds the minimum, new clusters are created or old ones are removed; otherwise, clusters are adjusted. This framework helps managers update customer segmentation with new information and enhance marketing strategies accordingly.

In (Luo et al., 2023), a new Bayesian nonparametric model named Hierarchical Fragmentation-Coagulation Processes (HFCP) is introduced for dynamic customer segmentation. This model works as follows: first, HFCP automatically determines the number of groups required to model diverse customer behavior. Next, the model can identify dynamic changes in customer behavior, such as the splitting and merging of groups. Using a hierarchical approach, HFCP discovers shared behavior patterns across different products. Additionally, HFCP outperforms previous models such as Homogeneous Poisson Processes (HomoPP), Non-Homogeneous Poisson Processes (NHPP), and Fragmentation-coagulation process)FCP(in predicting the purchase behavior of new customers and addresses overfitting issues. This model employs Fragmentation-Coagulation Processes to model changes in customer purchasing behavior and helps companies adjust their marketing strategies based on accurate behavioral patterns. Empirical results demonstrate that HFCP effectively models customer purchasing behavior and improves performance.

In (John et al., 2023), various clustering algorithms for customer segmentation in the UK online retail market were compared. The study used a UK-based online retail dataset and evaluated algorithms such as K-means, Gaussian Mixture Model (GMM), Density-Based Spatial Clustering of Applications with Noise (DBSCAN), agglomerative clustering, and Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH). The results indicated that the GMM algorithm achieved the best performance with a Silhouette Score of 0.80. This research demonstrates that advanced algorithms can improve the accuracy and efficiency of customer segmentation, allowing companies to fine-tune their marketing strategies and better understand customer purchasing behavior.

Table 1 is related to the summary of the review of the theoretical foundations of customer segmentation, in which the information provided in the literature review is summarized and the different types of segmentation models are given along with the studies conducted by researchers and the years associated with each method. From Tables 1 and it can be concluded simply and briefly that the previous researches on customer segmentation have used the method of new purchase, repeat purchase, and purchase amount.

Fig. 1 shows the algorithms used in the reviewed articles. Most of the reviewed articles have used unsupervised learning algorithms for customer segmentation. Therefore, unsupervised learning algorithms have been used in this research.

In Fig. 2, a comparison is made between the unsupervised learning algorithms used in the articles. The most popular algorithms are K-Means, fuzzy C-Means, Spectral, hierarchical, K-shape, self-organizing maps, concentration-based clustering and finally K-Medoids.

According to the literature review, it was concluded that unsupervised learning algorithms are mostly used in research. Unsupervised

Table 1
Summary of overview of the theoretical bases of customer segmentation.

	New purchase Repeat purchase Purchase amount	Neural network	Time series	Clustering	Classification	Regression
(Akhondzadeh-Noughabi & Albadvi, 2015; Parvaneh et al., 2014; Yanovitzky & VanLear, 2008)	✓					
(SARI et al., 2016)	✓					
(Khajvand & Tarokh, 2011)	✓					
(Khajvand et al., 2011)	✓					
(Heldt et al.; Anitha & Patil, 2019)		✓		✓		✓
(Daneshvar et al., 2020)	✓					
(Tavakoli et al., 2018)	✓				✓	
(ABBASIMEHR & BAHRINI, 2022)	✓				✓	
(Christy et al., 2018)					✓	
(Çavdar & Ferhatosmanoğlu, 2018)	✓					
(Emami & Derakhshan, 2015)				✓	✓	
(Sivaguru, 2023)	✓					
(Luo et al., 2023)						✓
(John et al., 2023)					✓	
(Batista et al., 2014)					✓	
(Arbelaitz et al., 2014)					✓	
(Dunn, 1973)			✓			
(Alboukaey et al., 2020)			✓			
(Montero-Manso & Hyndman, 2021)		✓				
(ABBASIMEHR & Shabani, 2021)		✓				
(Hamidi, 2016)		✓				

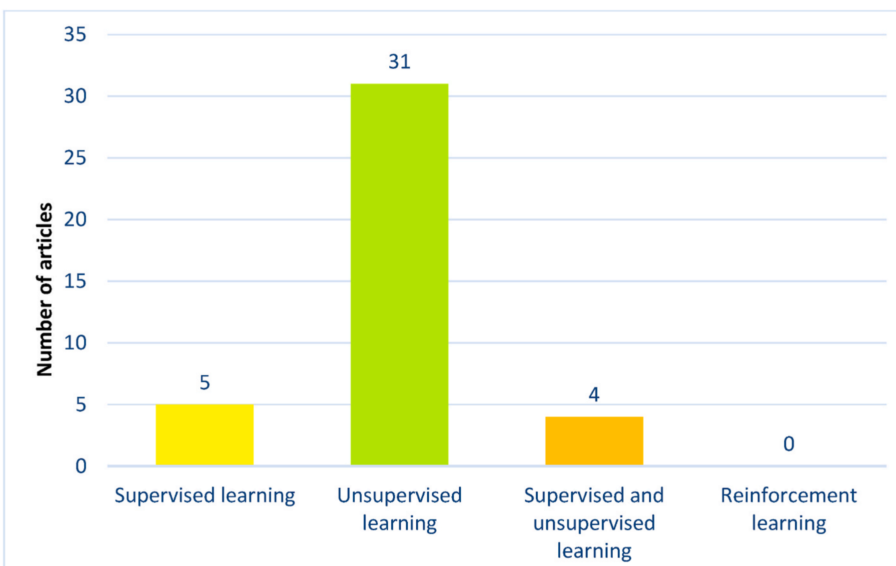

Fig. 1. Comparison chart of data mining methods used in articles.

learning algorithms are highly attractive to researchers due to capabilities such as data clustering and discovering hidden patterns in them. These algorithms are used especially when the data is not labeled and there is a need to segment and recognize patterns and relationships between the data. After examining more details among the unsupervised algorithms, the top four unsupervised algorithms were selected. This choice was made based on the importance, performance and potential of these algorithms in data clustering. Therefore, in this research, these four best algorithms (K-Means, Hierarchical Spectral, Fuzzy C-means) have been used to perform data clustering. This choice has been made according to the literature review and based on the ability and efficiency of these algorithms in the field of data clustering.

Reviewing past studies reveals that most customer segmentation research focuses on Recency, Frequency, and Monetary (RFM) values due to their simplicity and efficiency. However, these studies treat RFM features independently and assign equal weights, overlooking their interrelationships and varying importance. Additionally, static RFM-based methods fail to capture the dynamic nature of customer interactions, limiting their ability to adapt to evolving behaviors and market conditions. Moreover, customer lifecycle costs are frequently neglected despite their significance in understanding customer profitability. To address these gaps, this paper incorporates Customer Cost into the segmentation process, employs a genetic algorithm to optimize feature weights, and utilizes time series analysis to account for temporal dynamics. These enhancements result in more accurate and adaptive customer segmentation, providing a comprehensive understanding of customer behavior and supporting the development of effective, responsive marketing strategies.

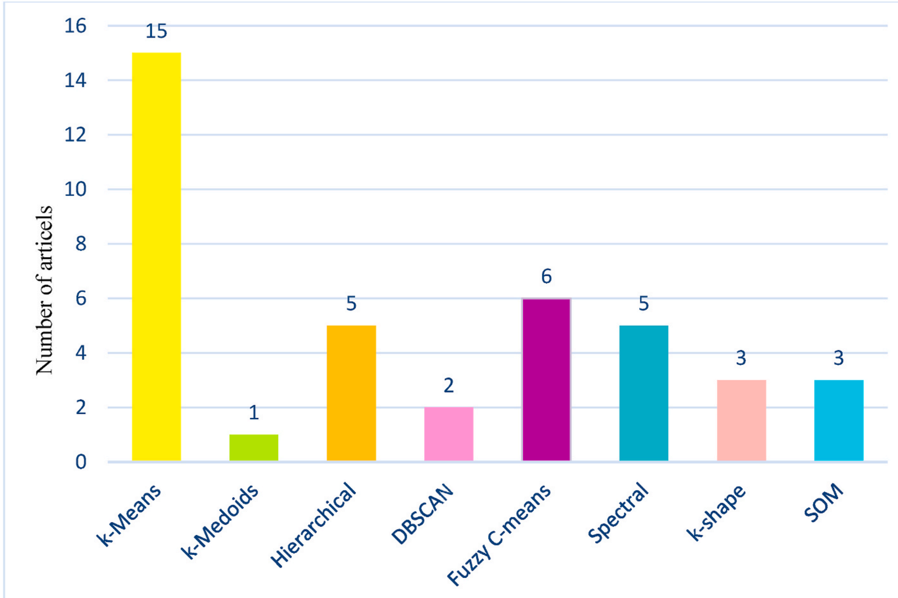

Fig. 2. Comparative diagram of unsupervised algorithms used in articles.

3. Research methodology

As shown in Fig. 3, the proposed model effectively integrates new transactions, the number of transactions, transaction profit, and cost with time series clustering. It extracts time series data from a stream of timestamped transactional data to depict the behavior of customers. The details of each step in the model are explained below.

3.1. CRISP methodology

CRISP is a common methodology in the field of data mining that is used to carry out data mining projects. This methodology consists of several steps that are executed sequentially and includes the complete process of data analysis. In the following, its steps are explained (ABBASIMEHR & SheikhBaghery, 2022).

3.1.1. Business understanding phase

The business understanding step in the CRISP methodology is related to the deep understanding that the researcher needs to know about the business under investigation. In this step, the researcher should examine and analyze the market, products and services, customers, competitors, organizational structure and other business-related factors. To get to know the business, different methods can be used, including:

Studying documents and related sources: By studying reports, articles, books and other sources related to business, one can get the necessary information about the type of activities, target market, competitors and other aspects of the business.

Interviewing experts: by interviewing people who work in the industry or organization under investigation, you can gather their opinions and knowledge about the business and reach a higher understanding.

Direct observation: By directly observing the activities, products and services of the business, you can get a better understanding of its performance and characteristics. As a result, by knowing the business, it is possible to consider the best approach and appropriate solutions for the successful implementation of the research and facilitate obtaining the desired results.

3.1.2. Data understanding phase

The step of data understanding in CRISP methodology is related to the process of understanding and analyzing data. In this step, the researcher must collect the available data and analyze them in a regular and organized manner. To understand the data, various methods and techniques can be used, including:

Data Collection: Business related data is collected. This data includes customer transaction data, information related to products or services, financial data and other business related information.

Data Preprocessing: In this step, the data is preprocessed. It includes data cleaning, removing incomplete or duplicate data, converting data format and structuring them properly.

Data analysis: Using data analysis methods, patterns and trends in the data can be identified. Various techniques such as descriptive statistics, data mining methods, modeling and other methods are used. By doing the mentioned steps, a better understanding of the data and business characteristics can be achieved, which will help in the subsequent analysis and modeling.

3.1.3. Data collection and preparation phase

The data collection and preparation step in the CRISP methodology belongs to the process in which the necessary data for subsequent analysis and modeling are collected, extracted, cleaned and prepared. In this step, steps should be taken to make the data useable and reliable. To collect and prepare data, the following methods and techniques can be used:

Data collection: In this step, data related to the research is collected. It includes customer transaction data, product or service data, financial data, historical data and other required information. Data can be collected from internal company sources (such as database management systems) or external sources (such as Internet sources or public data sources).

Data Cleansing: In this step, the data is cleaned and cleared of errors, mistakes and incomplete data. Methods such as removing duplicate data, compensating for missing data, converting data format, using data cleaning techniques and correcting invalid data can be used.

Selecting and extracting features: In this step, the important and required features for analysis and modeling are extracted from the data. These attributes can include numerical attributes such as mean, variance, and categorical attributes such as product type, customer gender, etc. The correct selection of features is very important in data analysis and can have a great impact on the accuracy and efficiency of prediction models.

Data transformation and preparation: In this step, data is prepared

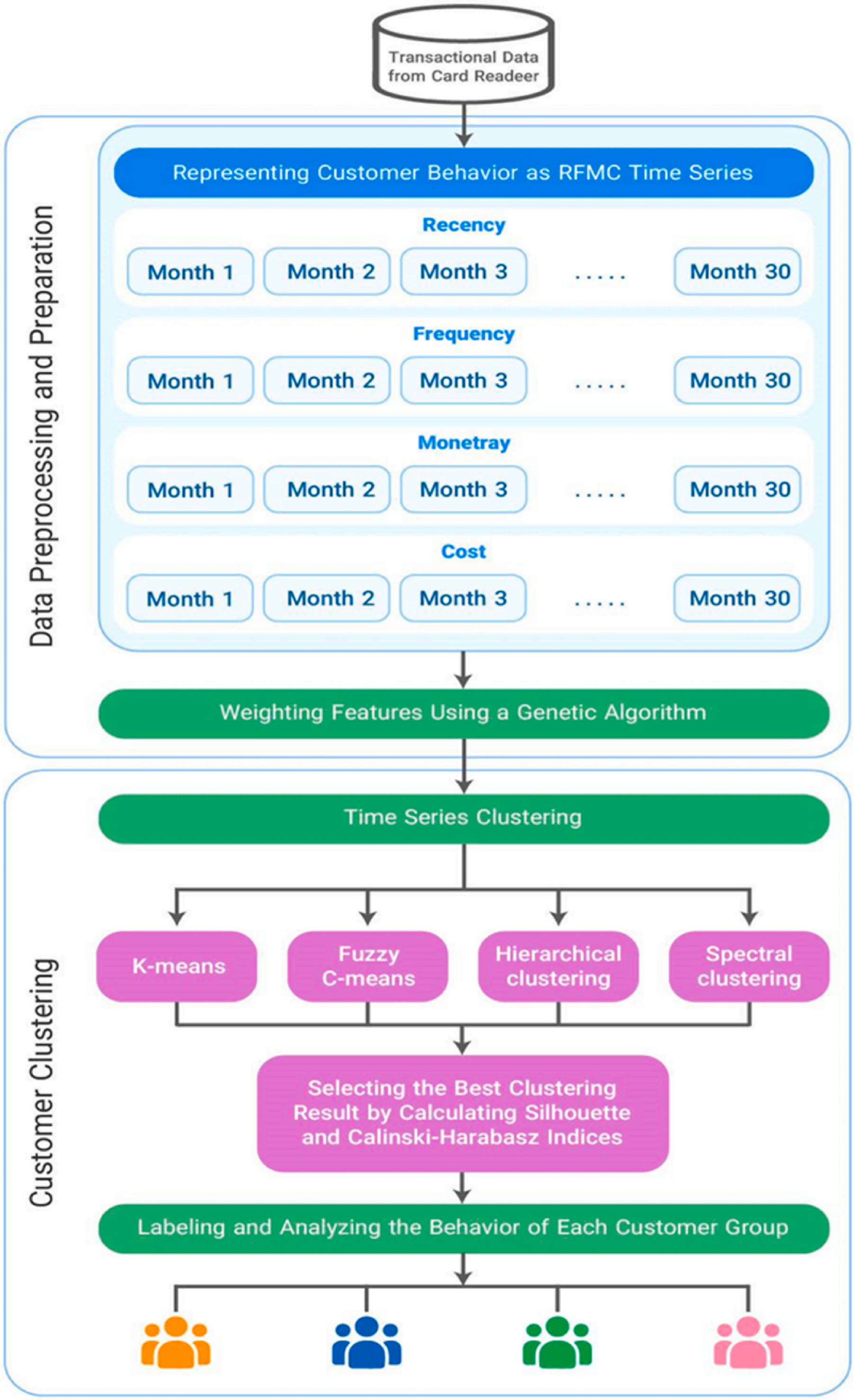

Fig. 3. Proposed research model.

for use in modeling. This includes converting the data into a suitable format for data mining algorithms, normalizing the data, reducing the dimensionality of the data, as well as dividing the data into two sets of training and testing. By performing the data collection and preparation steps correctly, the data will be ready to be used in modeling and analysis. This makes it more efficient to use the data and obtain more accurate and reliable results.

3.1.4. Modeling phase

The modeling phase in the CRISP methodology is the stage in which prediction, classification or clustering models are created from the collected and prepared data. The purpose of these models is to explain the patterns in the data and also to predict the behavior and events in the future. The main stages of the modeling phase are:

Selection of algorithms: In this step, based on the type of analysis desired (such as prediction, classification, or clustering) and the characteristics of the data, suitable algorithms are selected for modeling. In other words, the type of problem and the nature of the data will determine the choice of algorithms.

Building models: In this step, using the selected algorithms, different models are created on the data. These models can include statistical models such as regression and factor analysis, machine models and adaptive models such as clustering algorithms.

Training and evaluation of models: The built models are trained using training data and then evaluated using test data. Evaluation of models includes criteria of accuracy, correctness, recall and other related criteria. If needed, the models are improved and retrained by changing the parameters.

Selection of the final model: After evaluating the models and comparing their performances, the model that has the best performance and meets the analytical needs is selected as the final model. Using CRISP methodology in the modeling phase helps researchers to use data in a structured and step-by-step manner and to create acceptable and reliable models for data analysis.

3.1.5. The criterion phase of measurement and model evaluation

Model measurement and evaluation criteria in CRISP methodology are used to evaluate the performance of built models. These criteria are determined based on the problem under investigation and are usually used for prediction and classification models. Some of the commonly used criteria are as follows:

Accuracy: the ratio of the number of correct samples predicted by the model to the total number of samples.

Accuracy: The ratio of the number of real positive samples correctly identified by the model to the total number of positive samples predicted by the model.

Recall: The ratio of the number of true positive samples correctly identified by the model to the total number of positive samples in the data.

F-measure: A measure that is a combination of precision and recall and is used to balance the two measures.

Confusion matrix: a table that shows the number of correct and incorrect samples predicted by the model and is used as an evaluation tool in classification problems.

The area under the performance characteristic curve: This measure is used for classification models and indicates the ability of the model to distinguish between two different categories. Each of these criteria can be used to evaluate the performance of the models depending on the need and the problem under investigation. Also, by combining these criteria and using other criteria, a more comprehensive evaluation of the models can be done.

3.1.6. Deployment phase

The development phase in CRISP methodology is the last phase of data mining research. In this phase, the models built by the researchers in the previous phases are used and used for the required predictions and analyses. In this phase, after building the models and training them using the training data, the models are evaluated using the test or validation data. If the performance of the models is acceptable, they are used for use in the project or real applications. If the performance of the models does not reach the desired results, researchers may need to change the parameters, change the algorithms or use better and more suitable data to improve the performance of the models. Finally, after ensuring the performance of the models, they can be used for further predictions and analysis in projects and business decisions. As the final stage, this phase shows the results and benefits obtained from data mining research in real applications and plays a very important role in evaluating the effectiveness of research.

3.2. Representing customer behavior as recency, frequency, monetary, and cost (RFMC) time series

In this stage, transactional data of customers, timestamped, are aggregated based on monthly intervals. Time series related to each variable for the customer are calculated. The definition of time series for the variables of transaction recency, number of transactions, transaction profit, and customer cost are as follows.

- **Transaction Recency (t):** The number of months that have passed since the customer's last transaction in a specified time period.
- **Transaction Frequency (t):** The number of transactions conducted with the card reader during the specified time period.
- **Transaction Profit (t):** The amount of revenue generated from the customer's transactions during the specified time period.
- **Customer Lifecycle Costs (t):** The costs incurred by the organization for the services and products provided to the customer during the specified time period. These costs include expenses for card reader rolls, periodic visits, costs related to technical or consulting services, costs associated with the use of specific equipment and technologies, and other related service provision costs.

In this methodology, customer behavior analysis is performed considering four features: transaction recency, number of transactions, transaction profit, and customer cost. This comprehensive approach allows for the examination of the impact of each of these features on customer behavior.

3.3. Weighting features using a genetic algorithm

In this stage, a genetic algorithm is used to weight the features in time series clustering. The objective of this process is to find an optimal set of weights that appropriately assign importance to the features during clustering. Using these weights allows for more precise clustering.

3.4. Time series clustering

Time series clustering is an analytical method that allows for the grouping of similar time series into separate clusters. In this method, time series are first extracted as samples of temporal data. Then, using appropriate distance metrics, the distances between time series are calculated. These metrics can include Euclidean distance, Manhattan distance, and other similar measures based on the characteristics of the time series (Batista et al., 2014). Subsequently, four powerful clustering algorithms—hierarchical, spectral, K-means, and fuzzy C-means are employed for clustering.

3.5. Selecting the best clustering result by calculating Silhouette and Calinski-Harabasz indices

At this stage, after segmenting customers using various combinations of algorithms and feature weights, the quality of the segmentation models needs to be evaluated using appropriate clustering validity indices. Clustering is an unsupervised method that aims to divide data into segments with high internal similarity and low inter-segment similarity (Arbelaitz et al., 2014). Internal clustering validity indices examine how similar data points within each group are to each other, helping to select the best clustering result.

3.6. The main stages of the proposed methodology

3.6.1. Labeling and analyzing the behavior of each customer group

In this stage, each of the resulting clusters is labeled, and the behavior of each group is examined to identify their dominant patterns over time. To summarize the main steps of the proposed methodology, Fig. 3 has been drawn. This model receives customer transactions with timestamps as inputs and creates a time series of features for each customer, representing their behavior over time. Then, using a genetic algorithm, appropriate weights for each feature are extracted, and customer segmentation is performed through time series clustering. Specifically, various combinations of algorithms and feature.

Data mining is a process that uses various techniques and algorithms to extract patterns, trends and useful information from a large volume of data. This process involves examining data, identifying hidden patterns and connections that can be used in decision making, forecasting and optimization. The main goal of data mining is to discover hidden patterns and relationships that can be used to make decisions, make predictions, and improve performance. This process generally includes steps such as data preprocessing, feature selection, pattern discovery, and model evaluation. Data mining is used in various fields to obtain useful information and make decisions based on data (Sivaguru, 2023).

In this research, SQL Server software was used to extract data from the database. First, the required data was extracted from the database using SQL Server. Then, using the Python programming language and developing the corresponding codes, the data were processed and analyzed. This processing includes the use of Python libraries related to data mining.

In this research, various data mining techniques were used. These techniques include.

- Data preprocessing to prepare the data for the next steps.
- Extracting important features related to changes and time patterns.
- Extracting the proper weight of features using genetic algorithm.
- Data clustering using clustering algorithms.
- Evaluation of clusters using Silhouette and Kalinsky criteria.
- Analyzing customer behavior and labeling customers based on clusters and forming specific groups.

weights are explored to find the best clustering model according to the clustering validity index. Finally, the resulting clusters are analyzed to reveal their dominant patterns over time.

4. Empirical study

This section presents a real-world example of implementing the proposed framework using data from a banking payment service company. The proposed framework and all its components have been implemented using Python 3.7. The input data consists of 30 months of transactional data from card reader devices, including information such as card reader ID, transaction ID, date, and transaction amount. This dataset comprises 195,844,085 detailed purchase transactions by customers.

4.1. Data description

4.1.1. Software and implementation environment

In this research, two popular and powerful softwares, Python and structured query language, have been used to implement customer segmentation models. Python programming language has been used as the main language for data analysis and running algorithms. By having various libraries and useful capabilities for data mining, Python helps to analyze customer data and perform segmentation with high accuracy. Also, a structured query language database has been used to store and manage customer data. Using a structured query language, information about customers is stored in tables and used for data extraction and processing.

In this research, the database of a famous and large Iranian payment service provider company has been used. This data includes 195844058 transaction records of 48948 customers of this electronic payment company in the historical period of thirty months. Some of these data are shown in Table 2. Each row of this table represents a customer. There are 30 records for each customer for a 30-month period, considering that 30 months of each customer have been used in a time series. The profit that reaches the organization from each customer transaction is collected in the period of one month and is placed in the transaction profit column.

Table 2
Sample data used in the research.

Terminal number	Profit from the transaction	Number of transactions	Transaction recency	Customer cost
123456	247000	323	0	22000
678901	342000	409	2	123000
234567	987000	711	1	187000
890123	102000	167	5	145000

The number of transactions includes the number of customer transactions in a month, and the transaction freshness is the number of months that have passed since the last customer transaction. In the beginning, the transactional data of customers were small and detailed, but due to the large volume of data and the complexity of the processing operations, the data were analyzed in the form of monthly summaries. By collecting and aggregating data, monthly data can be used to check the behavior of each customer in a time series. In this process, four main features are used, including the number of transactions, transaction freshness, transaction amount, and customer cost. The customer cost feature is calculated from the amount of roll consumed, the cost of periodic visits and other services provided to the customer. Then, for each customer, the information related to the last thirty months from the four aforementioned features was extracted monthly and the necessary analyzes were performed on the data. It should be noted that all the characteristics of the customers have a value in the 30-month period under review, (Description of the data used in the research is shown in Table 3).

4.1.2. Data processing

Data processing is done to prepare and improve their quality before using them in the next steps. Regarding the data used in this research, due to the use of transaction data of banking customers, the data has been collected from a clean and complete database, so there is no need for special pre-processing because the data has been collected from a clean and correct database. But a normalization step has been applied to the data and a monthly summary of microtransactions has been done. This operation has been done in order to use the data more easily and optimally in the next steps.

4.1.3. Feature extraction

Important features have been extracted from the data. At this stage, meaningful and useful features have been extracted from the customer data set, which play an important role in describing and interpreting customer behavior. These attributes are usually determined based on transactional data including transaction count, transaction recency, transaction amount, and customer cost. Using these features, it is possible to identify customer patterns and behaviors. Extracting features from customer data is a key step in the process of analyzing and segmenting customers, which provides more possibilities and capabilities to interpret and predict the future behavior of customers. To extract the number of transactions feature, the number of transactions performed

Table 3
Description of the data used in the research.

Feature	Description of the feature used
Transaction recency	The number of months passed since the customer's last transaction within a month is checked, the lower the better.
Number of transactions	The number of transactions that a customer has reviewed within a month, the higher this feature is, the better.
Profit from the transaction	The amount of profit obtained from a customer's transactions in a period of one month, the higher this feature, the better.
Customer cost	The amount of money that the organization pays for a customer in a period of one month, this cost includes periodical visits, consumption roll and other services provided to the customer. The less this feature, the better.

by each customer in a certain period of time is considered. To extract the transaction recency feature, the elapsed time since the last transaction performed by each customer is considered. To extract the transaction profit feature, the total profit of all transactions performed by each customer in a certain period of time is calculated from the transaction amount. Finally, to derive the customer cost attribute, the total cost amount that the customer received for the services and products is calculated. Feature extraction has been done on customer data using calculations and operations appropriate to each feature. These features are then used in the next stages of customer segmentation to describe and interpret customer behavior.

4.1.4. Data normalization

The purpose of data normalization is to create conditions where features are equal based on their importance and impact in analyzing and segmenting customers. This work helps to facilitate the process of data interpretation and analysis and increases the accuracy and usability of the analysis results. This is in order to remove any deviations and differences in sizes between features and to create a balanced distribution of data. Outliers were removed before data normalization. The outlier data were removed in such a way that only those customers who had transactions in at least twenty months remained. Then, the data were normalized using the MinMax normalization method. In this method, the feature values are normalized to a certain interval, usually between 0 and 1. There are different methods for data normalization, but in this research, the min-max normalization method was used because the min-max method places the data values in a certain range, which made the data comparable and hence The impact of data drift on the final results is reduced. Also, by examining the literature review in the second chapter, it was observed that this normalization method is one of the most common normalization methods used in the articles. Therefore, according to the advantages and literature review, it was decided to use the Min-Max method for data normalization.

$$Min\_max\ normalization = \frac{X - Xmin}{Xmax - Xmin} \tag{1}$$

In this formula, X represents the initial value of the feature. Xmin is the minimum feature value in the data and Xmax is the maximum feature value in the data. Using this formula, the features are normalized between the interval [0, 1].

4.1.5. Evolutionary algorithms

Evolutionary algorithms include several sub-branches. One of the commonalities between these algorithms is that the input of each of these algorithms is a population of people. The pressure of the environment makes the most appropriate and compatible person with the environment to be selected as the final solution. For this purpose, a quality function is considered for each person in the population. The general goal of these algorithms is to increase the value of the quality function related to each person and select the person with the highest quality function as the most compatible person from the population. The higher the value of the quality function, the more compatible that person is with the surrounding environment. Based on this function, individuals are selected as parents to produce the next generation. The act of producing children is done by two operators, mutation and combination, which are applied to the parents. The act of compounding is an act that is applied to two parents and is created by those two children. The mutation operator is performed only on a parent and a child is produced by it. The two operators of combination and mutation cause the emergence of new people in the society. Now these new people will compete with the old people of the society to be in the next generation. that this competition is based on their quality function. This procedure is repeated until the person with the appropriate quality function is selected as a solution to the problem (Christy et al., 2018).

In evolutionary algorithms, a random optimization occurs, which is modeled on evolution in nature and two main heuristics are obtained in these algorithms.

- ***Survival of the fittest:*** It is important in the selection operator in such a way that the one who is the strongest has more chance of survival and has more possibility for mating.
- ***Recombination***: by combining in the answer, we can hope to get better answers. One of the characteristics of the evolutionary algorithm is that it is blind, so that it does not see its way, and the only thing it needs is to be given the possibility to evaluate its performance, because if it can evaluate its performance, it can find its way. Another feature is that it simplifies the problem and uses a series of codes to make decisions, which are generally binary, but there are other types as well (Emami & Derakhshan, 2015).

4.2. Representing customer behavior as RFMC time series

In this stage, the data was first normalized using min-max normalization. Then, the transactional data was aggregated into monthly intervals. Based on the definitions of the RFMC variables, the R, F, M, and C time series for each month were extracted. Since all selected customers have transactions in every month, the R variable will be zero for all customers. Therefore, the R variable is not considered in the analyses as it does not contribute to distinguishing customer segments. Additionally, the transaction frequency of each customer indicates the number of transactions made by each customer. It is also evident that the monetary value of each transaction can vary. Thus, a high number of transactions does not necessarily equate to high monetary value. For this reason, the monetary variable is considered as the profit from the transaction rather than the transaction amount for analyzing customer behavior. The ultimate goal of any company is to achieve optimal profitability. In customer segmentation, the cost feature is also considered. In the examined sample, which includes card reader transactions, the costs Incurred by customers during their relationship with the organization are taken into account. These costs include expenses for card reader rolls, periodic maintenance visits, card reader malfunctions, and other costs borne by the organization for its customers. By considering cost in customer segmentation, groups of customers with similar cost patterns can be identified, allowing for optimal strategies to manage costs and enhance organizational efficiency.

4.3. Weighting features using a genetic algorithm

In the feature weighting stage, a genetic algorithm is used to determine the appropriate weights for the features. The genetic algorithm is a computational method inspired by the mechanisms of evolution and natural selection in nature. Using this algorithm, suitable weights for each feature are extracted to optimize their impact on customer analysis and segmentation. In this stage, after data preparation, the genetic algorithm is applied. The genetic algorithm uses an evolutionary process to identify the best weight for each feature. Initially, a population of weights is generated. Then, through the use of crossover and mutation operators, subsequent generations are created. In each generation, the weights are improved and adjusted based on an evaluation function to meet the optimal weight for each feature (Dunn, 1973).

After running the genetic algorithm and gradually improving the weights, the best weight values for each feature are extracted. This method uses a linear combination of features, meaning that the features are combined with different weights to obtain a final value for customer analysis and segmentation. The permissible values for these features range between [−1, 1], meaning that the weights for the features are between one and negative one. The values associated with the parameters of the genetic algorithm used are presented in Table 4.

The genetic algorithm was implemented using four clustering algorithms: hierarchical, spectral, K-means, and fuzzy C-means, resulting in four output charts corresponding to each fitness function. Figs. 4–7

Table 4
Describes the genetic algorithm optimization parameters.

Parameter Name	Parameter Description	Value
Maximum Number of Generations	The maximum number of generations that the genetic algorithm should run. After reaching this number of generations, the optimization operation stops.	100
Number of Individuals per Generation	The number of people (population) in each generation from the genetic algorithm,	200
Mutation Probability	The probability of performing mutation operations on each gene in each mutation generation means the random change of one bit of the gene at a certain point.	0.1
Elitism Ratio	A proportion of the population that is passed on to the next generation as superior individuals (elite) and is excluded from the operations of combination and mutation.	0.01
Crossover Probability	Probability of combining operations on two individuals from the population. Fusion means combining different parts of two people to create a new person.	0.5
Selection Ratio	A proportion of the population used to select a parent for inclusion in the next generation.	0.3
Type of Crossover	The type of composition used in the composition operation may be uniform composition or other types.	uniform
Maximum Number of Generations without Improvement	The maximum number of generations during which no improvement in optimization has been made. This parameter can be useful to stop the algorithm if there is no improvement.	None

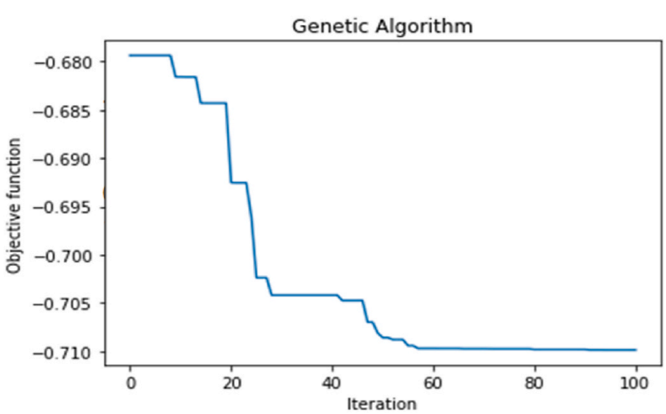

Fig. 4. Output of the genetic algorithm with K-means fitness function.

represent the performance of the genetic algorithm in segmenting customers using each of these algorithms. Based on the outputs obtained from these charts, the spectral clustering algorithm performed with higher accuracy compared to the other algorithms. Therefore, to improve clustering accuracy, the spectral algorithm and the weights provided by it were used. These results indicate that the genetic algorithm, with the fitness function calculated by the spectral algorithm, offers greater capability in customer segmentation, contributing to the increased accuracy and efficiency of the segmentation process. Additionally, the best solution found by the genetic algorithm for weighting features is as follows: the weights for the three features—number of transactions, transaction profit, and cost—are −0.70, 0.80, and 0.90, respectively, and the objective function is 0.91 for the spectral algorithm. In subsequent steps, the weights are multiplied by the features, and clustering is performed with the new values.

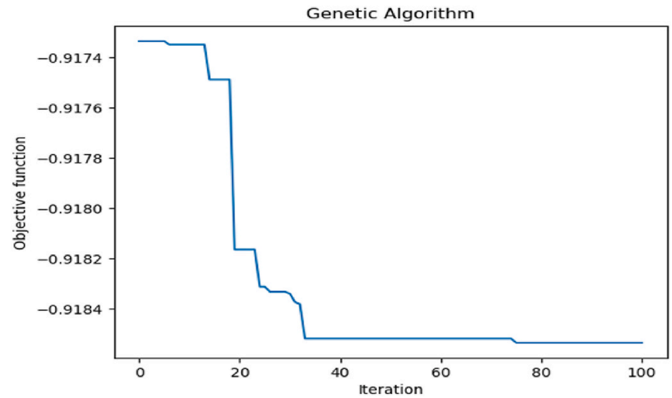

Fig. 5. Output of the genetic algorithm with spectral fitness function.

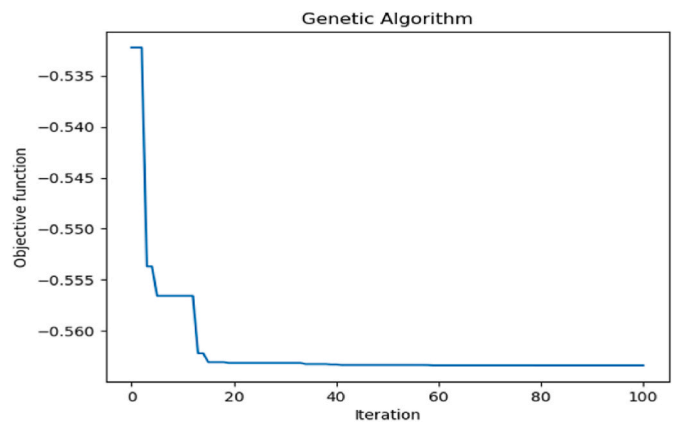

Fig. 6. Output of the genetic algorithm with fuzzy C-means fitness function.

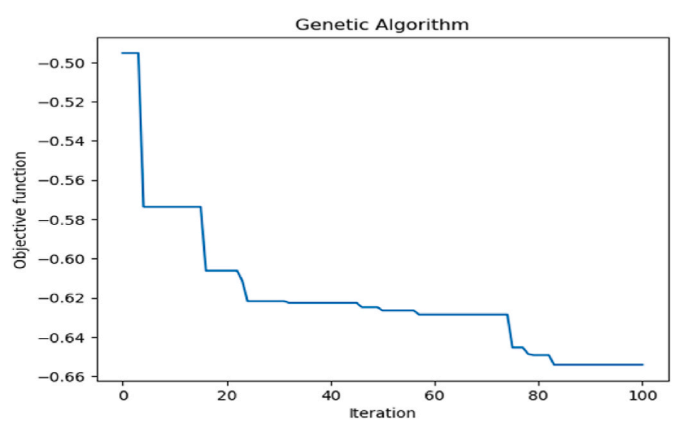

Fig. 7. Output of the genetic algorithm with hierarchical fitness function.

The general and comparative explanations of these charts are briefly presented in Fig. 8 so that the results and evaluations are clearly understandable.

By using the genetic algorithm and the fitness function calculated by the Spectral algorithm, the best accuracy for customer segmentation was obtained. To compare the results, a bar graph has been drawn that shows the segmentation accuracy using different algorithms (K-Means, hierarchical and fuzzy C-Means) along with the Spectral algorithm. The bar chart shows that by using the genetic algorithm and the spectral fit function, the accuracy of the segmentation has been improved and they have provided a significant improvement compared to other algorithms. This means that the genetic algorithm with the fitness function

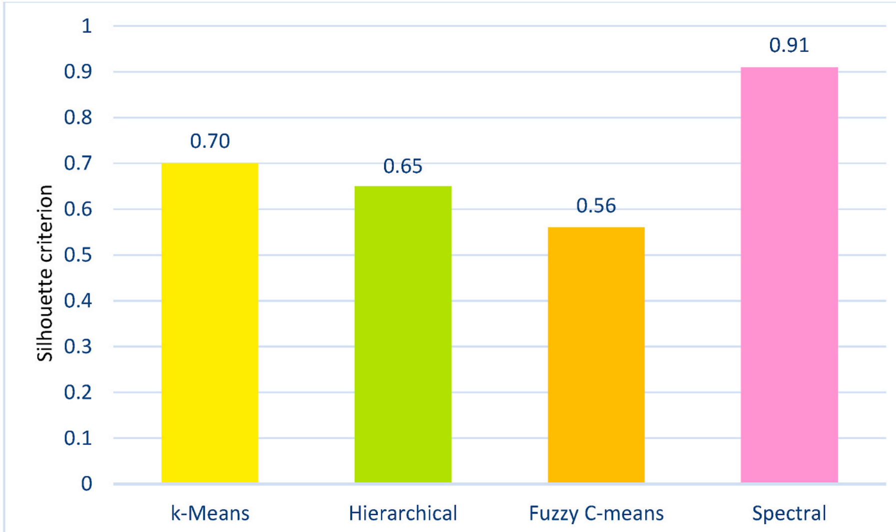

Fig. 8. The comparative diagram of the implementation of the genetic algorithm with different fitness functions.

calculated by the Spectral algorithm provides the best ability to segment customers and increases the accuracy and efficiency of the segmentation process. The best solution found by the genetic algorithm is as follows:

Weights: [-0.70162901, 0.80858289,0.901110443]

and the value of the objective function: 0.91.

The genetic algorithm has been able to provide a high-accuracy clustering model by optimally assigning the weights. The value of Don's index obtained indicates a very good fit of the clustering with the data and a suitable separability between the clusters. This means that the optimal weights obtained by the genetic algorithm well represent the features in customer clustering and the resulting clustering model provides more accurate results. Then, using these optimal weights, clustering algorithms were implemented and customers were segmented. These optimal weights have managed to adjust and select the best features for clustering customers. The result of clustering with these weights can be seen in the clustering section. Using these results, it can be claimed that the genetic algorithm with selected parameters has succeeded in finding the optimal weights for customer clustering, and the resulting clustering is more accurate and better than before.

4.4. Time series clustering

After normalizing the data and weighting the features using a genetic algorithm, the next step is customer segmentation using four popular algorithms identified in the literature review. The prepared data is segmented using spectral, hierarchical, fuzzy C-means, and K-means algorithms, and the best algorithm is identified based on the Silhouette and Calinski-Harabasz indices.

The optimal number of clusters for customer segmentation is determined using the Dunn index. The Dunn index is a quantitative measure that determines the optimal number of clusters based on the distance between clusters and within clusters. By calculating this index for different numbers of clusters, the optimal number of clusters is obtained. The higher the Dunn index value, the better the clustering results (Alboukaey et al., 2020). The different Dunn index values calculated are shown in Table 5. Additionally, the changes in the Dunn index are illustrated in Fig. 9, where the highest Dunn index value is obtained for three clusters.

Table 5
Dunn index calculation results.

Optimal Number of Clusters	Dunn Index
2	0.4190
3	0.4411
4	0.3983
5	0.4257
6	0.4268
7	0.4006
8	0.3997
9	0.3987

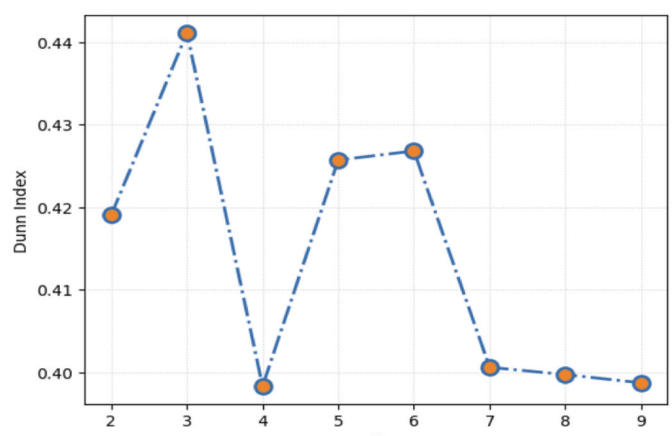

Fig. 9. Dunn index variation chart based on the number of clusters.

4.5. Selecting the best clustering result by calculating Silhouette and Calinski-Harabasz indices

In this stage, the accuracy of clustering is improved using a genetic algorithm. Initially, before applying the genetic algorithm, clustering is performed using four algorithms: spectral, hierarchical, fuzzy C- means, and K-means. Then, after executing the genetic algorithm with optimized weights, another round of clustering is conducted using the same four algorithms.

The quality of clustering is evaluated using the Silhouette and Calinski-Harabasz indices. This comparative analysis shows that the genetic algorithm with optimized weights brings significant improvement in clustering accuracy.

4.5.1. Segmentation using K-Means algorithm

In customer segmentation using K-Means algorithm, a method called K-Means is used to divide customers into different groups. In this algorithm, first a number of primary centers (cluster centers) are determined and customers are assigned to the closest cluster center. The cluster centers are then updated based on the average number of customers assigned to them, and the customers are reassigned to the nearest cluster center. This process is repeated repeatedly until a stable state is established and the cluster centers do not change. The K-Means algorithm has advantages such as simplicity and relatively high speed compared to other algorithms for data segmentation. This algorithm is generally used in clustering problems and is considered as one of the most used and popular algorithms in this field (Christy et al., 2018). The results of customer segmentation using the K-Means algorithm on the analyzed data are shown in Table 6.

4.5.2. Segmentation using C-Means fuzzy algorithm

Fuzzy C-Means (FCM) algorithm is a clustering algorithm used for data segmentation using a fuzzy approach. In this algorithm, each data is probabilistically assigned to one or more clusters, instead of being explicitly assigned to a cluster. In the C-Means fuzzy algorithm, the centers of the clusters are initialized randomly. Then, for each data, the probability of belonging to each cluster is calculated using the concept of fuzzy membership. These membership probabilities are then weighted to update the cluster centers. This process is repeated until the changes in the cluster centers are less than a threshold value. The advantages of C-Means fuzzy algorithm include the ability to model fuzzy data patterns, the ability to simultaneously assign to several clusters, and the ability to apply it to data with a complex structure (Hamidi & Vafaei, 2009). The results of customer segmentation using the C-Means fuzzy algorithm on the analyzed data are shown in Table 7.

4.5.3. Segmentation using spectral clustering algorithm

The spectral clustering algorithm is a clustering algorithm based on the spectral analysis of graphical information from the data. In order to segment the data, this algorithm uses the information of the graph structure and places the data in different clusters based on the spectral characteristics of the graph. The performance of the spectral clustering algorithm is that first a graph is created for the data. Then the graph spectrum is calculated and clustering is done using the graph spectrum. For this purpose, first the eigenvectors corresponding to the smallest values of the graph spectrum are extracted and then these vectors are used as input for the clustering algorithm, such as the K-Means algorithm. The main advantage of spectral clustering algorithm is in modeling and clustering data with complex structure. This algorithm can identify hidden patterns in the data and carry out accurate clustering according to the structural information of the data (Luo et al., 2023). The results of customer segmentation using the spectral clustering algorithm on the examined data are shown in Table 8.

4.5.4. Segmentation using hierarchical algorithm

Hierarchical algorithm is a clustering algorithm based on the formation of a hierarchy of clusters. In this algorithm, first each point is considered as a separate cluster. Then, using the distance criteria between the points, the clusters are ranked and hierarchically connected to each other to finally reach a single large cluster. Hierarchical algorithm works in such a way that in each step, two closer clusters are combined with each other and become a larger cluster. This combination of clusters is done based on the criteria of the distance between the clusters. Distance measures may include Euclidean distance, Manhattan distance, or any other distance measure chosen based on the properties of the data and the clustering problem. The main advantage of the hierarchical algorithm is the ability to represent the hierarchy of clusters. This algorithm is able to divide points into different clusters and also display the hierarchical structure between these clusters (Christy et al., 2018; Çavdar & Ferhatosmanoğlu, 2018). The results of customer segmentation using the hierarchical algorithm on the analyzed data are shown in Table 9.

Table 6
K-Means algorithm implementation results before and after genetic algorithm weights are applied.

Evaluation criteria	Before genetic algorithm	After genetic algorithm
Silhouette index	0.44	0.67
Calinski index	8608	15574

Table 7
Results of C-Means fuzzy algorithm implementation before and after applying genetic algorithm weights.

Evaluation criteria	Before genetic algorithm	After genetic algorithm
Silhouette index	**0.43**	0.67
Calinski index	8523	16884

Table 8
Results of spectral clustering algorithm implementation before and after applying genetic algorithm weights.

Evaluation criteria	Before genetic algorithm	After genetic algorithm
Silhouette index	**0.43**	0.91
Calinski index	7596	17307

4.6. Comparison of customer segmentation models

The comparison of customer segmentation models using genetic algorithm and without using genetic algorithm has been discussed. In order to check more precisely which model performed best in customer data analysis and to what extent the genetic algorithm was effective in improving clustering. All customer segmentation algorithms that were introduced in the previous sections are compared with each other and the best segmentation model is determined using two criteria, Silhouette and Kalinsky. To compare segmentation models, two criteria, Silhouette and Kalinsky, have been used. The silhouette measure measures the degree of separation and integration between clusters. This measure is calculated based on the distance between samples within the cluster and the distance between samples in adjacent clusters. The silhouette value for each cluster ranges from −1 to 1, with higher values indicating better separation between clusters (Khajvand et al., 2011).

$$\text{Silhouette}(i) = \frac{b(i) - a(i)}{\max\,(a(i), b(i))} \tag{2}$$

b: indicates the minimum average distance between points from a cluster that are not clustered to other points from adjacent clusters.
a: represents the average internal distances of the points of each cluster to the reference point (the central point of the cluster).

Calinski's criterion measures the number of clusters and the quality of separation between clusters. This measure is calculated based on the difference between the internal average of the clusters and the average

Table 9
Results of the implementation of the hierarchical algorithm before and after applying the weights of the genetic algorithm.

Evaluation criteria	Before genetic algorithm	After genetic algorithm
Silhouette index	**0.39**	0.66
Calinski index	7360	15373

between the clusters. A high value of this criterion indicates that the clusters are separate and the segmentation quality is better (ABBASIMEHR & BAHRINI, 2022).

$$Calinski\_Harabasz = \frac{b}{w} + \frac{N-k}{k-1} \quad (3)$$

bindicates the interclusterity of the data and is calculated as the sum of the squares of the distances between the central points of the clusters and the central point of the entire data. w indicates that the data are within a group and in sum

The square of data distances is calculated from the center of the cluster to which they belong. N represents the total number of data points. K represents the number of clusters.

Figs. 10–13 compare the Silhouette and Calinski-Harabasz indices for the different algorithms before and after executing the genetic algorithm. These charts illustrate a marked increase in the values of the Silhouette and Calinski-Harabasz indices for all algorithms after optimization by the genetic algorithm. This increase indicates better cluster segmentation after the genetic algorithm's optimization. The results of the Silhouette and Calinski-Harabasz indices show that the spectral clustering algorithm performed with higher accuracy compared to the other algorithms. Therefore, for further improvement and increased clustering accuracy, customer segmentation was continued using the spectral algorithm.

According to the values of Silhouette and Calinski criteria for each algorithm and comparing them with each other, the best segmentation model for customers is the Spectral algorithm because it has the highest Silhouette and Calinski values.

4.6.1. Customer segmentation using genetic algorithm

By using genetic algorithm, optimal weights were obtained for the attributes of transaction number, transaction profit and customer cost. Then, using these optimal weights, the clustering algorithms were re-run with the same previous conditions, and it is shown in Figs. 12 and 13 that the clustering models with optimal weights have improved compared to the models without using optimal weights, and the accuracy and efficiency of clustering have improved.

4.7. The results of the implementation of the spectral algorithm with the optimal weights of the genetic algorithm

According to the values of silhouette and Kalinsky criteria for each algorithm and comparing them with each other, the best segmentation model for customers using genetic algorithm weights is Spectral because it has the highest silhouette and Kalinsky values compared to other algorithms. For this reason, customer segmentation has been done using the Spectral algorithm and the optimal weights of the genetic algorithm, and the results are shown in Table 10.

In Table 10, the number of samples in each cluster is shown. Each row of this table represents a cluster and each column represents the number of samples in the same cluster. This information tells us how many customers each cluster contains and how large it is in terms of number of instances.

In Table 11, a statistical summary of the data in each cluster is shown. These characteristics can include mean, variance, median, minimum value, maximum value, etc. This information shows us how the customers of each cluster performed on various attributes (such as number of transactions, transaction profit, and customer cost) and whether these customers differ from each other on these attributes. Also, this table can provide information about the distribution of data in each cluster, and this information can be used in data analysis and review.

Fig. 14 shows the graph of changes in the average values of all three characteristics of transaction profit(a), transaction number(b) and cost (c) based on clustering groups. In this diagram, each clustering group is displayed with a specific color. The x-axis of the graph shows the dates, and the y-axis of the graph shows the average attribute values for each group on each date. By looking at this graph, you can see the changes in the average values of the features for each of the groups over time. This graph shows how the clustering groups' performance has changed over time and whether their average attribute values have increased or decreased. Also, by comparing different groups, it can be seen which group had the best performance in terms of the average values of the features. In which periods of time there were visible changes and this information was used in the analysis and review of customer data.

4.8. Labeling each customer segment and analyzing the behavior of each group

This section of the paper analyzes different customer groups. By calculating the center of each group, the temporal information of that segment is identified. Figs. 15–17 show line charts of the average values

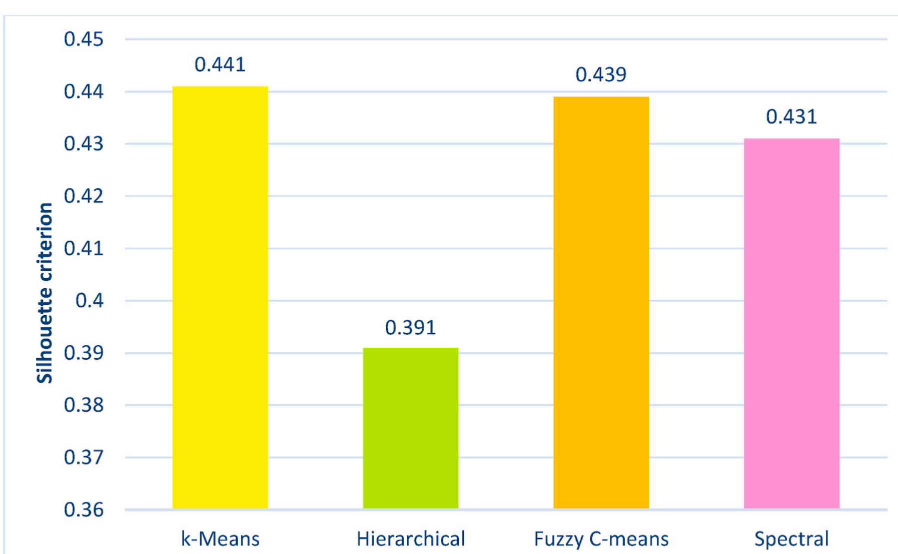

Fig. 10. Comparison chart of accuracy of clustering algorithms with Silhouette criterion without genetic algorithm.

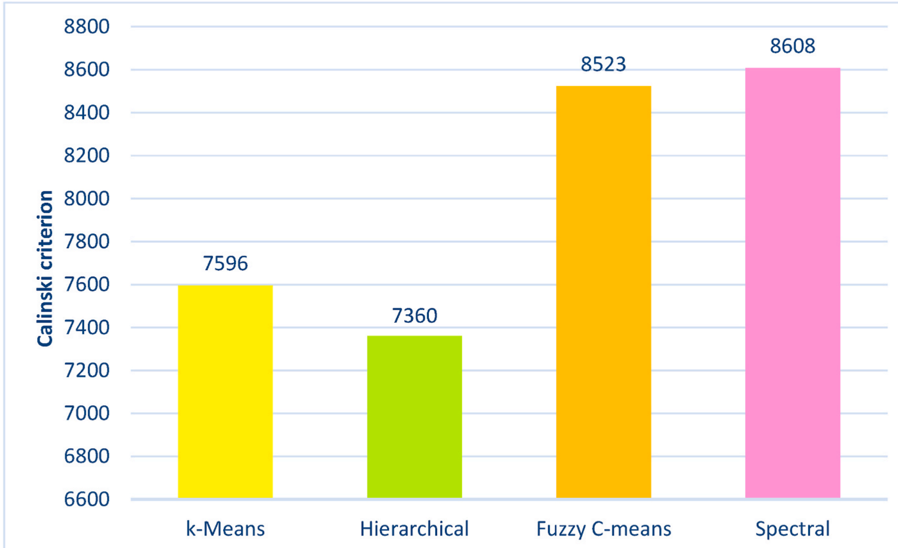

Fig. 11. Comparison chart of accuracy of clustering algorithms with Calinski criterion without genetic algorithm.

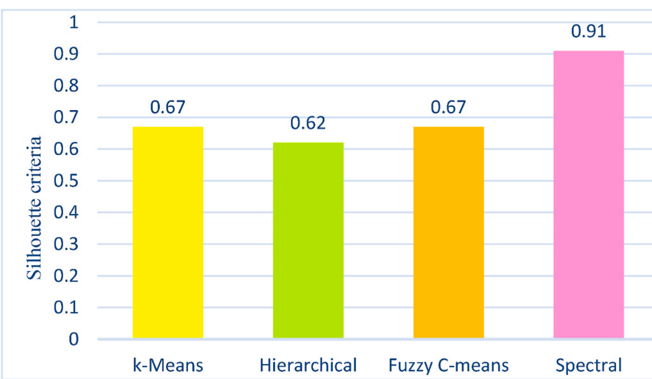

Fig. 12. Comparison chart of the accuracy of clustering algorithms with the Silhouette criterion with the genetic algorithm.

for the three features: transaction profit, number of transactions, and cost, based on the clustering groups. In these charts, each feature transaction profit, number of transactions, and cost is displayed in a specific color. The x-axis represents the dates, and the y-axis represents the average values of the features for each group on each date. By observing these charts, one can see the changes in the average values of the features for each group over time. These charts illustrate how the performance of the clustering groups has changed over time and whether the average values of their features have increased or decreased. Additionally, by comparing different groups, one can determine which group has had the best performance in terms of average feature values, observe notable changes in certain time periods, and utilize this information for analyzing and reviewing customer data.

Fig. 18 displays the values of transaction features, profit, and cost for each of the clustered groups. In this chart, each column corresponds to a feature, and for each feature, 3 bars are drawn for groups one, two, and three, respectively. Each group is also distinguished by a different color. The values shown in this chart represent the average values for each feature for each group. This chart generally shows how each group differs in terms of the examined features compared to the other groups. Based on this statistical information, it can be concluded that Group 3 has a higher average number of transactions and transaction profit compared to the other two groups. Therefore, this group consists of more active customers compared to the other two groups. Following Group 3, Group 1 has a higher share of transactions and profit overall. Based on these two features, Group 2 ranks third, as customers in this group have the lowest share of transactions and profit compared to the other groups. On the other hand, it is observed that the cost for customers in Group 2 is lower than in the other two groups. This result indicates that Group 2 is the best group in terms of cost. After Group 2, Group 3 comes next, followed by Group 1. This point shows that, on average, customers in Group 2 conduct their transactions at a lower cost compared to the other two groups. Table 12 shows the behavioral analysis of each customer group. By analyzing these features and the associated statistical values for each group, one can better understand the differences and advantages of each group in the business, leading to better decision-making regarding strategies and customer retention. Additionally, time is a crucial factor in this analysis. According to Figs. 15–17, which show the average values by group, it can be observed that customers in Group 3 have exhibited consistent and balanced behavior over the 30-month period studied. This indicates their loyalty. Customers in Group 2 had high profits and transactions in the initial months, but after some time, these values decreased, indicating that they are on the verge of being lost. This point highlights the need to adopt effective strategies to retain Group 2 customers and prevent their loss. Moreover, customers in Group 1 had few transactions initially but have recently shown significant growth and have become profitable customers. Overall, by analyzing this statistical data and charts, one can better understand the differences and behaviors of each group and implement appropriate strategies to make necessary improvements and changes in the business.

5. Performance evaluation

This section evaluates the performance of the proposed framework, divided into three subsections explained below.

5.1. Innovations of the research

One of the most significant innovations of this paper is that, unlike most previous methods that considered equal weights for the parameters of new purchase, purchase frequency, and purchase amount, this study uses a genetic algorithm to obtain optimal weights for customer features. In this model, by optimizing and adjusting weights using a genetic algorithm, various features are appropriately utilized in customer segmentation.

This research considers the cost feature in customer segmentation. In the examined sample, which includes card reader transactions, the costs incurred by customers during their relationship with the organization

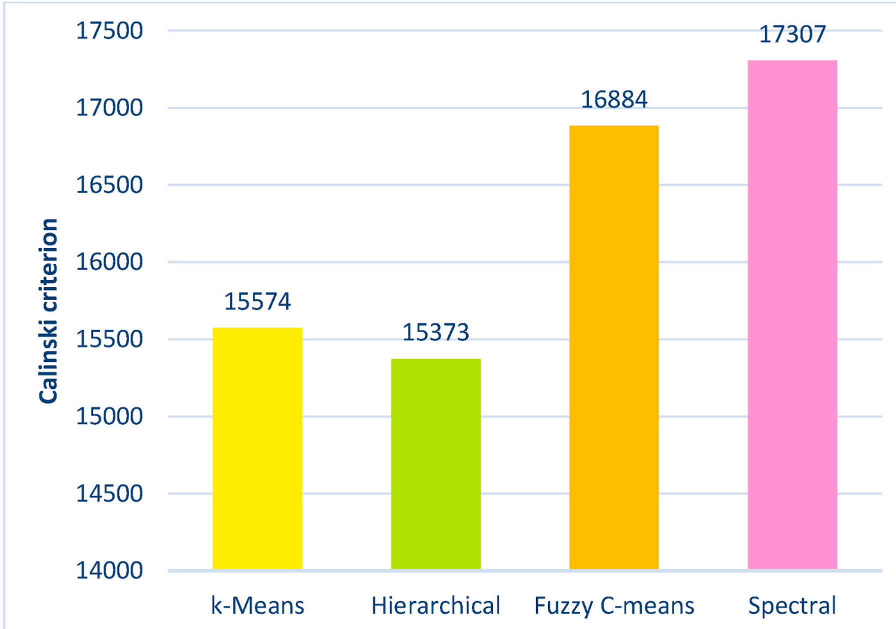

Fig. 13. Comparison chart of the accuracy of clustering algorithms with Calinski criterion and genetic algorithm.

Table 10
The number of samples in each cluster of customers for k = 3.

Cluster number	Number of cluster samples
0	**15513**
1	**3557**
2	**1814**

Table 11
Statistical summary of data in each cluster.

		Number of transactions	Transaction Profit	Customer Cost
Second group	**Count**	30	30	30
	mean	0.58	0.64	0.55
	std	0.3	0.31	0.3
	min	0	0	0
	25%	0.42	0.51	0.43
	50%	0.62	0.73	0.60
	75%	0.81	0.87	0.75
	max	1	1	1
		Number of transactions	**Transaction Profit**	Customer Cost
First group	**Count**	30	30	30
	mean	0.40	0.40	0.43
	std	0.32	0.32	0.34
	min	0	0	0
	25%	0.09	0.10	0.12
	50%	0.34	0.37	0.42
	75%	0.69	0.72	0.69
	max	1	1	1
		Number of transactions	**Transaction Profit**	Customer Cost
Zero group	**Count**	30	30	30
	mean	0.48	0.49	0.56
	std	0.33	0.34	0.40
	min	0	0	0
	25%	0.11	0.10	0.06
	50%	0.55	0.59	0.78
	75%	0.75	0.77	0.92
	max	1	1	1

are taken into account. These costs include expenses for transaction rolls, periodic maintenance visits, card reader malfunctions, and other costs borne by the organization for its customers. By considering the cost feature in customer segmentation, the best strategies for cost management and resource optimization can be found. Analyzing customer costs and their impact on the organization's revenue and profit can help develop strategies to reduce costs. Moreover, it is possible to identify customer groups with similar cost patterns and provide optimal strategies for managing organizational costs. The customer segmentation model developed in this paper has been successfully implemented in an electronic payment company, delivering better results than the previous segmentation model used by the company. Thus, this model is commercially viable and applicable in real-world scenarios.

5.2. *Integrating the proposed framework into an information system for improved decision making*

Some of the applications that can be implemented based on this proposed approach include.

1. **Customer Segmentation**: different behavioral groups of customers can be analyzed over time.
2. **Customer Churn Detection**: Identifying customers who are likely to churn and taking appropriate actions to retain them is essential (Alboukaey et al., 2020). Recognizing individuals who intend to leave the customer list and taking suitable measures to retain them is crucial. most studies in the field of churn detection are based on static approaches.
3. **Customer Behavior Prediction**: Predicting customer behavior is critical in customer relationship management (Montero-Manso & Hyndman, 2021). Robust time series forecasting methods can become a valuable tool for managers. The main difference between this approach and previous research is its exploitation of the predictability feature.

5.3. *Research limitations*

The first limitation of the research was processing large volumes of data. This study required strong hardware resources to process and analyze the extensive data. The second limitation is that only customer

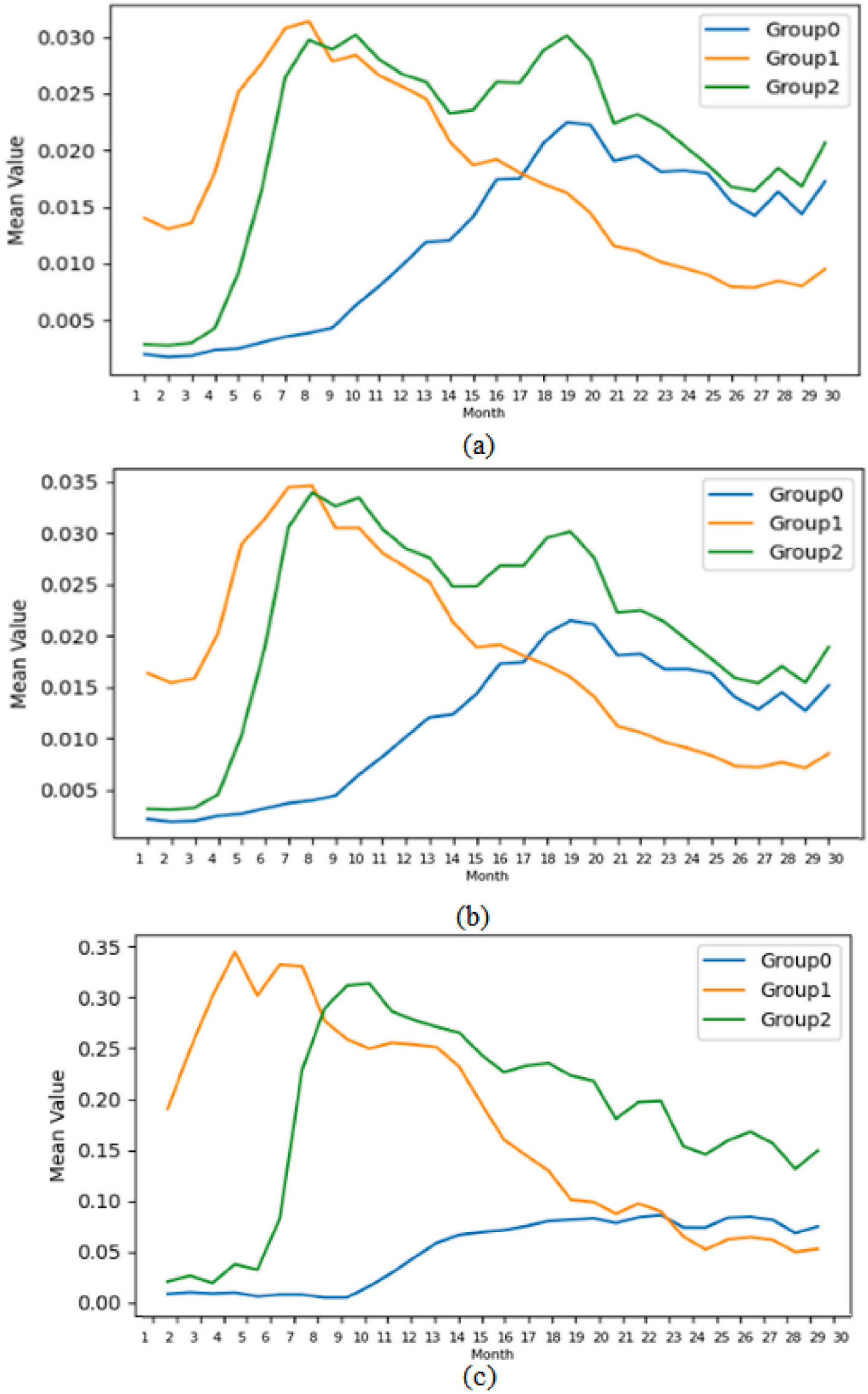

Fig. 14. The linear graph of the average values of the features by features for each group.

transaction data was available; other information such as location, store size, and number of staff were not accessible for a more detailed analysis of the segments obtained.

6. Conclusion

In this paper, a customer segmentation model for dynamic segmentation based on modeling customer behavior has been presented. Customer segmentation has been carried out by considering the entire customer lifetime from inception to the present, looking not only at past behavior but also anticipating future behavior. Therefore, this research fully accounts for the customer lifetime in segmentation. This study offers a dynamic model for customer segmentation, capable of monitoring changes in customer behavior through time series clustering. This shift from static to dynamic segmentation enables continuous improvement and adaptation to customer behavior. The study considers new purchases, purchase frequency, and purchase amount as a linear combination, without ignoring the relationships between these features. Using a genetic algorithm, optimal weights were assigned to customer features, resulting in improved segmentation compared to previous

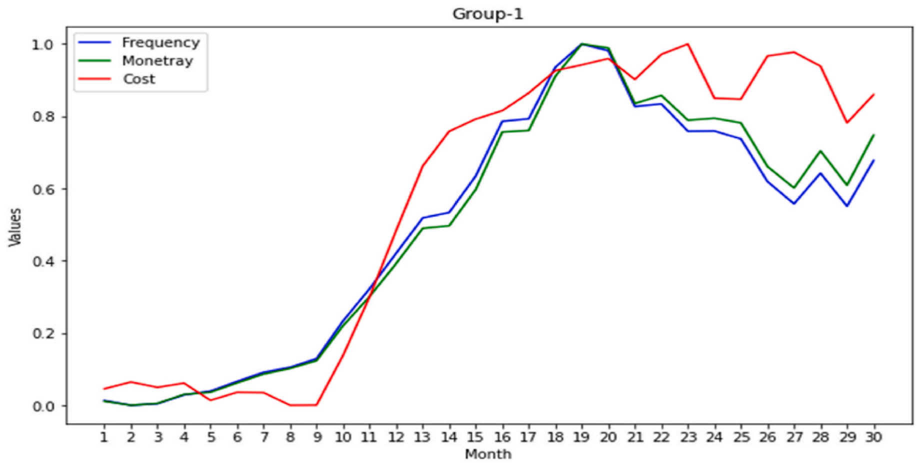

Fig. 15. Line Chart of the Average Values of Features for the first Cluster.

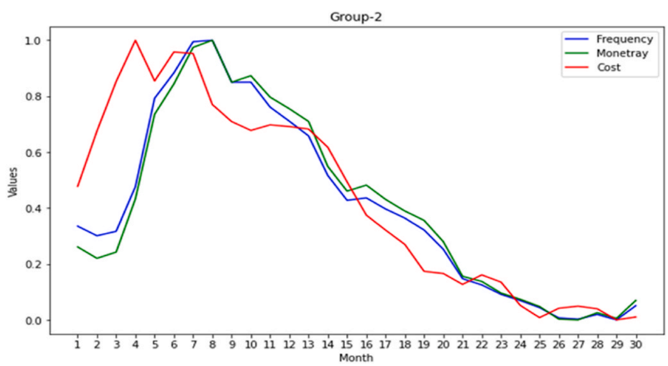

Fig. 16. Line Chart of the Average Values of Features for The second cluster.

models. The innovations of this research include modeling customer segmentation based on behavioral changes over the customer lifetime, using time series analysis, considering cost features in segmentation, and the ability to continually improve segmentation based on behavioral changes. The model has been successfully implemented in an electronic payment company, yielding better results than the previous segmentation model. The genetic algorithm, by optimally weighting features, has provided a highly accurate clustering model. The obtained Dunn index indicates a very good fit of the clustering to the data and appropriate separability between clusters. This means that the optimal weights obtained by the genetic algorithm effectively represent the features in customer clustering, providing more accurate results. These weights were used to execute clustering algorithms and perform customer segmentation, successfully selecting and optimizing the best features for clustering. The results demonstrate that the genetic algorithm with the selected parameters successfully found optimal weights for customer clustering, resulting in more accurate and improved segmentation. In this study, a hybrid and efficient method for customer segmentation was achieved using the genetic algorithm and weight optimization. As shown in Table 13, the segmentation accuracy significantly improved compared to other methods. Additionally, the analysis revealed that the large data volume (195,844,085 transactions and 48,489 customers) provided comprehensive and extensive insights, resulting in more precise and reliable customer segmentation. Another distinguishing feature of this work is the inclusion of the cost feature in customer segmentation, unlike other studies that did not consider this feature. This research used a linear combination of features and analyzed customer data over 30 months, while similar studies considered periods of seven to eleven months. Therefore, given the improved segmentation accuracy, large data volume, inclusion of the cost feature, and extended analysis period, the results of this study are successful and effective, showing significant improvement over reference papers.

For **future research**, the scope of the data can be expanded to include other customer variables such as geographic location, age,

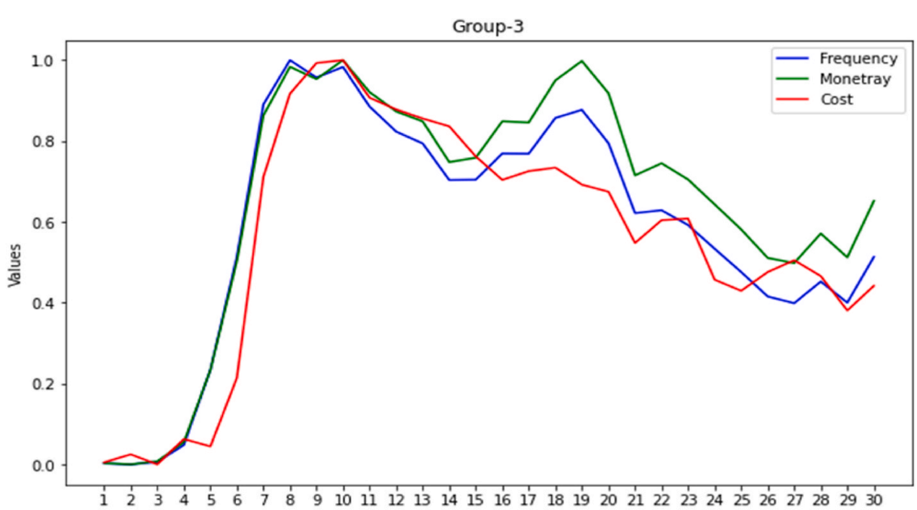

Fig. 17. Line Chart of the Average Values of Features for the third cluster.

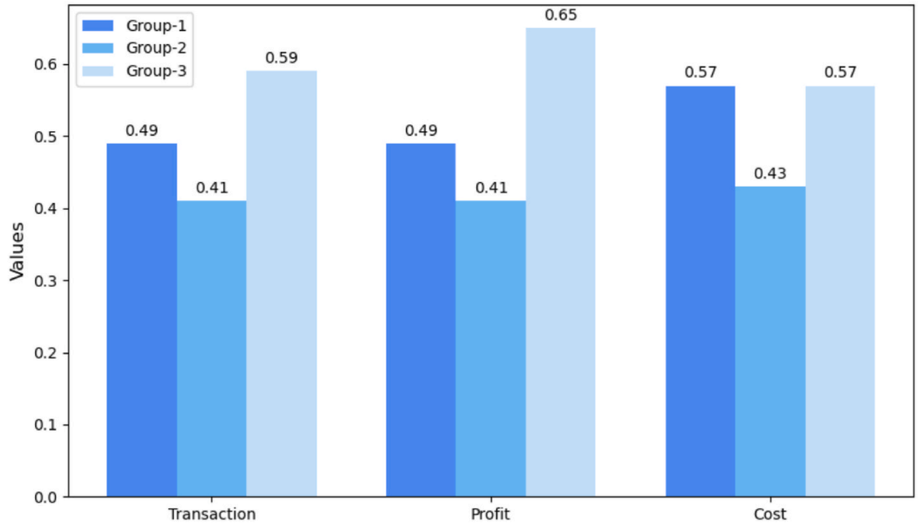

Fig. 18. Comparative chart of average feature values for each customer group.

Table 12
Behavioral analysis of each customer group.

Label	Size	Segment	Behavioral Analysis
Silver	15,513	Cluster One	Customers in this group initially had low profits but recently experienced significant growth and became profitable. Strategies should be devised to encourage these customers to engage in more transactions. Bank-centric strategies should focus on enhancing customer experience and satisfaction for this group.
Bronze	3557	Cluster Two	Customers in this group initially had high profits and transactions, but over time, these features decreased, indicating they are at risk of being lost. This signals the need for effective strategies to retain these customers and prevent their attrition.
Gold	1814	Cluster Three	Known as the golden customers, this group includes the most profitable customers due to their consistent high revenue generation. This group shows stable and loyal behavior. Strategies should be designed by the bank to maintain and enhance their loyalty and satisfaction, considering their significant value to the organization.

Table 13
Comparison of the present study with previous works.

Article	Algorithm	Data Volume	Study Field	Silhouette Index Value
ABBASIMEHR and BAHRINI (2022)	Spectral, Hierarchical, K-shape	2,156,394 transactions	Bank	0.46
ABBASIMEHR and Shabani (2021)	Hierarchical	259,000 transactions	Bank	0.46
Hamidi (2016)	K-Medoids, K-Means, Fuzzy C-means, Self-Organizing Map	123,684 customers	Bank	0.62
Current Study	K-Means, Fuzzy C-means, Spectral, Hierarchical	195,844,058 transactions	Bank	**0.91**

gender, and other features. Expanding the data scope will allow for more comprehensive analysis and more detailed examinations of customer behavior. Additionally, significant improvements in predicting customer behavior can be achieved using deep learning methods such as deep neural networks. These methods can identify more complex customer behavior patterns and enhance prediction accuracy. As indicated in this study, using evolutionary algorithms like the genetic algorithm can significantly improve feature extraction efficiency. Therefore, one of the latest members of the evolutionary algorithm family, Cartesian Genetic Programming (CGP), is introduced. Given the improved results in various papers using CGP compared to those using traditional genetic programming, it is suggested that feature combinations be performed using CGP. Continuing research based on the results of this paper can improve various aspects of the customer segmentation model and provide optimal strategies for enhancing customer relationships and increasing their lifetime value.

CRediT authorship contribution statement

Hodjat (Hojatollah) Hamidi: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Bahare Haghi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Availability of data and materials

The data used to support the findings of this study are available from the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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